09B — Deep-Dive Feature Audit: Combat Engine & Tactical Simulation (Part 2 of 6)

Spatial Battlefield Geometry, `NPC` vs `foe` Movement Asymmetry, The `24`-Tile Telepathic Row-Bodyblock Bug, Broken `evade`/`blink` Repositioning & AoE Wall-Clipping

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Physically Coherent Wuxia Battlefield**. Positions `(x, y)`, weapon reach, walls, spear-wall Zones of Control, lightness backsteps, and bodyguard dashes must obey the exact same continuous physics for both Sect Disciples and Enemies!
- **Master Roadmap for Category 09 (`09A–09F`)**:
  - `09A`: Core Damage Formula (`dmg = hpMax(d) * frac`), 12-Stat Split-Brain, Elemental Matchups & Behavioral Mastery Traits
  - **`09B` (This Document)**: **Spatial Battlefield Geometry, `NPC` vs `foe` Movement Asymmetry, `24`-Tile `ROW_BODYBLOCK` Bug, Sworn-Enemy Interception, Broken `evade`/`blink` Repositioning & AoE Selectors**
  - `09C`: Action Economy, Cast/Windup/Channeling/Poise-Break, Statuses, DoT Self-Harm/Reflect Loops, Acupoints & Boss Anatomy Breaker
  - `09D`: Multi-Entity Units — Summons, Shadow Clones, Constructs, Pet Assists, Gate Beasts, Weapon Avatars, Ancestral Phantoms & Party Synergies
  - `09E`: Fight Modes, Stakes, Morale/Yield/Retreat, 7-Layer Cheat-Death Stacking, Raid Core-Breach AI, Self-Defense Jail Bug & Heavenly Tribulations
  - `09F`: Grand Finale — Holistic Overview of the Entire Combat Ecosystem

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1. `[CMB-SPAT-01]` Critical `NPC` vs `foe` Movement Physics Split (`js/combat.js:228–252`): Sect Disciples Ignore `zocSlowMult`, Ignore Combat `move` (`slow`/`chill`/`haste`), & Snap to Integer Tiles!

Inspect `moveToward(W, u, tx, ty, run, fOpt)` in `js/combat.js:209–252` verbatim:

```javascript
if (u.foe) {
  const sp = (u.speed || 0.6) * (run ? 1.0 : 0.8) * clamp(ST.get(u).move / 100,
0.35, 2.0) * zocSlowMult;
  ... // Smooth sub-tile vector movement!
} else {
  const gx = Math.floor(tx), gy = Math.floor(ty);
  if (!u.path || (((W.t + (u.n || 0)) % 6 === 0) && (u._pathGx !== gx || u._pathGy
!== gy))) {
    u._pathGx = gx; u._pathGy = gy; AI.goTo(W, u, gx, gy);
  }
  AI.stepMove(W, u); // <-- COMPLETELY IGNORES ST.get(u).move AND zocSlowMult!!
}
```

Four Cascading Battlefield Bugs:

1. **`zocSlowMult = 0.52` (`Zone-of-Control` Spear-Wall Slow) Is `100%` Discarded on Sect Disciples**:
   - Line `209` computes `zocSlowMult = 0.52`, then line `250` calls `AI.stepMove(W, u)` without applying `zocSlowMult`! In Duels, Tournaments, Leadership Challenges, and Raids, sect NPCs sprint right past enemy Tanks/Polearms at full speed!
2. **Combat `move` Stat (`slow`, `chill`, `gravity_slow`, `haste`, `swiftWind`, `SHOES`) Does `0%` to Sect NPCs in Combat**:
   - `AI.stepMove` uses out-of-combat world-map `walkSpeed` instead of `ST.get(u).move`! Freezing or slowing a sect disciple in a Tournament or Raid never slows their combat movement!
3. **Sub-Tile Precision Loss & 6-Tick Path Lag (`Math.floor(tx), Math.floor(ty)`)**:
   - Truncating `(50.9, 50.9)` to `(50, 50)` causes melee disciples at short range (`dist < 3.5` with clear LoS!) to step toward tile centers rather than the opponent's true sub-tile position, and waits up to `6` ticks to update A* when an enemy kites!
4. **Fix**: When `!u.foe` is in active `'fight'` state (`f.state === 'fight'`):
   - Scale `AI.stepMove` step distance by `clamp(ST.get(u).move / 100, 0.35, 2.0) * zocSlowMult`!
   - When `dist <= 4.0` and `C.checkLineOfSight(W, f, u, target, { requiresLoS: true }).terrainBlocked === false`, use **direct sub-tile vector stepping** (identical to `u.foe` with `!MAP.blocked` slide!) instead of coarse 6-tick integer A*, falling back to A* only when `terrainBlocked === true` or `dist > 4.0`!

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2. [CMB-SPAT-02] The 24.0-Tile Telepathic ROW_BODYBLOCK Bug (js/combat.js:365, 1437–1484) & Sworn-Enemy GOLD_THREAD Interception (lines 378, 405–439)!

2.1 Telepathic Melee Bodyblocking Across Half the Sect Map (js/combat.js:1456)

Inspect C.resolveRowBodyblock (lines 1455–1484), which runs **first** (line 365, ahead of LOS_COVER and BODYGUARD_INTERCEPT!):

```
JavaScript
if (Number.isFinite(u.x) && Number.isFinite(d.x) && C.unitDist(u, d) > 24.0)
return false;
if (Number.isFinite(u.x) && Number.isFinite(a.x) && C.unitDist(u, a) > 24.0)
return false;
```

- **Severe Tactical & Visual Bug**:
  - 24.0 tiles is **384 pixels** (from the Sect Gate almost to the Meditation Hall!).
  - When an enemy assassin flanks 20 tiles behind the frontline and stabs an isolated BACK_ROW healer at dist = 1.1, a FRONT_ROW warrior standing **23 tiles away** telepathically absorbs the stab **without moving (blk.x, blk.y)**, and line 983 (a.castFx.x = actualT.x) snaps the assassin's melee slash VFX 23 tiles across the map!
  - Furthermore, because resolveRowBodyblock runs *before* BODYGUARD_INTERCEPT (<= 2.85 tiles), it steals interceptions from the actual nearby bodyguard!
- **Exact Fix**: Run BODYGUARD_INTERCEPT (<= 2.85 tiles) and LOS_COVER **before** ROW_BODYBLOCK, tighten ROW_BODYBLOCK when live coordinates exist to **C.unitDist(u, d) <= 3.0 && C.unitDist(u, a) <= 3.4**, and dash blk into cover position via C.projectSafePosition!

2.2 Mortal Enemies Take Bullets for Each Other & Auto-Bind GOLD_THREAD (lines 378, 405–439)

- resolveDefensiveInterception adds +1.0 for positive rel.a >= 30, **but applies zero penalty for BLACK_THREAD vengeance or bitter hatred (rel.a <= -30 / rel.v >= 45)**!
- A TANK (+1.8) under vanguardInterceptBonus (+1.25 $\to$ sc = 3.05 >= 1.35) leaps in front of a lethal strike to save their mortal enemy, and line 439 binds **GOLD_THREAD (ด้ายทองผู้มีพระคุณช่วยชีวิต)** on top of their BLACK_THREAD!

- **Fix**: Disqualify `al` from voluntary `BODYGUARD_INTERCEPT` (and skip `GOLD_THREAD` binding) whenever `BLACK_THREAD` links `al` and `d` or `rel.a < -25 || rel.v >= 45`!

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3. `[CMB-SPAT-03]` Early `return` in `evade` & `blink` (`js/combat.js:814–818, 888–892`) Prevents Archers From Stepping Back & Sets `_closedOnBy` Backwards!

Inspect `useSkill` lines `814–818` vs lines `925–936` verbatim:

```
JavaScript
// Line 814:
if (p.arch === 'evade') {
  ST.addStatus(a, 'evadeUp', 14, null, W.t);
  if (p.grantBuff && ST.applyNamedBuff) ST.applyNamedBuff(a, p.grantBuff, buffDur,
W.t, { casterId: a.id });
  if (d) { d._closedOnBy = a.id; a._gapClosedOn = d.id; } // <-- BACKWARDS FOR
RANGED ESCAPE!
  return; // <-- EXITS BEFORE LINE 933 (isRetreat 1.6-tile backstep) AND BEFORE
onCast!!
}
```

- **Triple Bug**:
  1. **Zero Physical Retreat**: An Archer/Zither-player cornered at `dist = 0.8` casts an `evade` escape art, hits `return` at line `817`, **never reaches line `933` (`isRetreat` `1.6`-tile backstep)**, and stays glued at `dist = 0.8`!
  2. **Inverted `_closedOnBy` State**: Line `816` sets `d._closedOnBy = a.id` while **leaving `a._closedOnBy = d.id` (`and aTac.closedOnBy = d.id`) intact**! Even after casting their Lightness Escape art, the Archer remains flagged `"ถูกประชิดตัว"` (`×0.92 dmg, -8 acc`)!
  3. **Pure `blink` Skills (`M <= 0`, `line 892`) Never Teleport**: Line `892` returns *before* line `926` (`blink` coordinate projection), leaving `(a.x, a.y)` unchanged!
- **Exact Fix**: Move the `Gap-Closer / Blink` (`lines 925–931`) and `Disengage / Retreat` (`lines 933–936`) coordinate-repositioning block **above** the `evade` (`line 814`) and `blink` (`line 888`) early returns, and in `evade`, if `pa.reachClass === 'RANGED_KITE' || mSpec.isRetreat`, clear `aTac.closedOnBy = null; a._closedOnBy = null` and backstep `1.6` tiles away from `d`! Also uncache `p.range` when `u._activeStance` changes (`skills.js:161` / `03A`) so `THUNDER_STEP_STANCE` (`+0.55` range) actually takes effect!

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4. [CMB-SPAT-04] Unbounded 120-Tile Ricochet Bounces & Wall-Clipping AoEs in COMPOSABLE_ENGINE.selectTargets (js/combat.js:1138–1190)!

1. **RICOCHET_CHAIN & SPLASH_NEIGHBORS Bounce Across the Entire 240×192 Map (line 1188)**:
   - `return sortByDistToOrigin(enemies, ox, oy).slice(0, limit)` has **zero distance filter**! If Enemy #1 dies at the South Gate `(118, 130)` and Enemy #2 is `90` tiles away at the North Wall, `SPLASH_NEIGHBORS` / `RICOCHET_CHAIN` arcs `90` tiles across the entire compound! Add `filter(u => C.unitDist(d || a, u) <= (sel === 'RICOCHET_CHAIN' ? 4.5 : 2.8))`!
2. **CONE & PIERCE_LINE Shoot Straight Through Solid Stone Walls (lines 1143, 1179)**:
   - Filter `coneHits` and `lineTargets` with `!MAP.blocked` raycast (`C.checkLineOfSight(W, f, a, u, { requiresLoS: true }).terrainBlocked === false`) unless `mSpec.ignoresObstacles`!
3. **ALL_ENEMIES Truncated to 2.0 Tiles on Melee Weapons (line 1165)**:
   - For explicit battlefield-wide ultimates (`sel === 'ALL_ENEMIES'`), use `const maxRange = Math.max(6.5, (mSpec && mSpec.max) || 6.5)` rather than `2.0`!

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5. Mandatory Verification Suite Additions (09B)

1. **Symmetric Disciple move & zocSlowMult Test**: Apply `slow` (`-35%` move) and enemy Tank ZoC (`0.52×`) to a sect NPC in `f.state === 'fight'` and assert per-tick displacement drops proportionally!
2. **evade Ranged Disengage & ROW_BODYBLOCK Proximity Test**: Have a melee warrior close on an Archer (`dist = 0.8`), cast an `evade` retreat skill, and assert `dist > 1.8` and `aTac.closedOnBy === null`; place a `FRONT_ROW` ally `15` tiles away and assert `resolveRowBodyblock` returns `null`.
3. **Sworn-Enemy Non-Interception Test**: Bind `BLACK_THREAD` between Tank A and Caster B; assert Tank A refuses voluntary `BODYGUARD_INTERCEPT` for Caster B and never auto-binds `GOLD_THREAD`.